Sayram 2D: Numerical Algorithm

Xin Tao

October 20, 2026

1 The 2D diffusion equation to be solved

Sayram 2D aims to solve the 2D diffusion equation with a loss (or source term) in the following form:

$$\frac{\partial f}{\partial t} = \frac{1}{G}\frac{\partial}{\partial Q_i}\left(GD_{Q_iQ_j}\frac{\partial f}{\partial Q_j}\right) - \frac{f}{\tau}. \tag{1}$$

Equivalently,

$$\frac{\partial f}{\partial t} = \frac{1}{G}\nabla\cdot(\Lambda\cdot\nabla f) - \frac{f}{\tau}. \tag{2}$$

where

$$\Lambda = \begin{bmatrix} GD_{Q_1Q_1} & GD_{Q_1Q_2} \\ GD_{Q_1Q_2} & GD_{Q_2Q_2} \end{bmatrix} \tag{3}$$

is the diffusion tensor. The equation is subject to the following Dirichlet (Γ_D) and Neumann (Γ_N) boundary conditions:

$$f = g_D, \quad \text{on } \Gamma_D, \tag{4}$$

$$q_N = \boldsymbol{n}\cdot(-\Lambda\cdot\nabla f), \quad \text{on } \Gamma_N. \tag{5}$$

The q_N is the flux satisfying $q_N \le 0$. Here $G > 0$ is the Jacobian of the coordinate transformation, the $D_{Q_iQ_j}$ are diffusion coefficients that may depend on time, and τ characterizes loss or source. The f/τ term can be easily handled, and we will mainly deal with the negative solution problem caused by the mixed diffusion coefficients using a positivity-preserving finite volume method of Gao and Wu (2015).

2 A positivity-preserving finite volume method

The PPFV method was developed by Gao and Wu (2015) and is reviewed here. Denoting the value of f at cell K as f_K, the basic principle of constructing the PPFV method is to write the equation to obtain $f_K(t+\Delta t)$ from $f(t)$ as follows:

$$f_K(t+\Delta t) = \sum_i \alpha_i f_{K,i}(t), \tag{6}$$

where $f_{K,i}$ is the value of f of the i_{th} cell that contributes to updating the value of f_K. Note that this summation may also include f_K itself. If all coefficients (α_i's) satisfy $\alpha_i \ge 0$, and $f_{K,i}(t) \ge 0$, then the above equation guarantees that $f_K(t+\Delta t) \ge 0$. Clearly, for the radiation belt diffusion problem, the initial phase space density for every cell is non-negative. This illustrates the positivity principle; the method below achieves it through an M-matrix.

We consider the diffusion equation part of Equation (2):

$$G\frac{\partial f}{\partial t} = \nabla\cdot(\Lambda\nabla f), \tag{7}$$

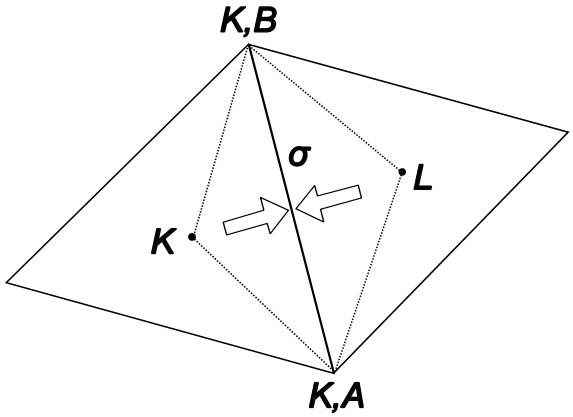


Figure 1: Illustration of the Nonlinear Two-Point Flux Approximation (NTPFA). We first construct a one-sided flux, from cell center K to edge σ, using only values of f_K, $f_{K,A}$, $f_{K,B}$. This process is then repeated for the cell L. The flux used in the PPFV method to update f_K or f_L is derived using the one-sided fluxes.

Integrating the above equation over a cell K, we have

$$\int_K G\frac{\partial f}{\partial t}\mathrm{d}V = \oint_{\partial K} \Lambda\nabla f \cdot \mathrm{d}\boldsymbol{A} \approx -\sum_{\sigma} \tilde{J}_{K,\sigma}, \tag{8}$$

where the summation is over all edges of the cell K. In the following, we demonstrate the construction of $\tilde{J}_{K,\sigma}$ based on the idea of NTPFA and Gao and Wu (2015), ensuring the positivity of f.

Consider cell K, as illustrated by Figure 1. Gao and Wu (2015) first constructed the one-sided flux from cell center K to the edge σ of the following form:

$$J_{K,\sigma} = \alpha_{K,\sigma,A}(f_K - f_{K,A}) + \alpha_{K,\sigma,B}(f_K - f_{K,B}) \tag{9}$$

$$= (\alpha_{K,\sigma,A} + \alpha_{K,\sigma,B})f_K - (\alpha_{K,\sigma,A}f_{K,A} + \alpha_{K,\sigma,B}f_{K,B}), \tag{10}$$

where $\alpha_{K,\sigma,A}$ and $\alpha_{K,\sigma,B}$ represent the coefficients to be determined. This $J_{K,\sigma}$ approximates

$$-\int_{\sigma} (\Lambda \cdot \nabla f) \cdot \boldsymbol{n}_{K,\sigma}\mathrm{d}s \tag{11}$$

using information from cell K only.

Gao and Wu (2015) demonstrate that, to ensure the final scheme is positivity-preserving, the only condition required is that for every cell such as K, the coefficients satisfy the condition

$$\alpha_{K,\sigma,A} + \alpha_{K,\sigma,B} > 0. \tag{12}$$

Gao and Wu (2015) proposed the following expressions for the coefficients,

$$\alpha_{K,\sigma,A} = |\sigma|\frac{\boldsymbol{n}_{K,\sigma}^T\Lambda_K\mathcal{R}(\boldsymbol{x}_{K,B} - \boldsymbol{x}_K)}{(\boldsymbol{x}_{K,A} - \boldsymbol{x}_K)^T\mathcal{R}(\boldsymbol{x}_{K,B} - \boldsymbol{x}_K)}, \tag{13}$$

$$\alpha_{K,\sigma,B} = |\sigma|\frac{\boldsymbol{n}_{K,\sigma}^T\Lambda_K\mathcal{R}(\boldsymbol{x}_{K,A} - \boldsymbol{x}_K)}{(\boldsymbol{x}_{K,B} - \boldsymbol{x}_K)^T\mathcal{R}(\boldsymbol{x}_{K,A} - \boldsymbol{x}_K)}. \tag{14}$$

Here $|\sigma|$ represents the length of the edge σ, $\boldsymbol{n}_{K,\sigma}$ is the outer unit column vector normal to σ from center cell K, and Λ_K is the diffusion coefficient matrix evaluated at the cell center K. The superscript T denotes the transpose of a vector or matrix. Column vectors $\boldsymbol{x}_K, \boldsymbol{x}_{K,A}$, and $\boldsymbol{x}_{K,B}$ represent the coordinates of the cell center, and vertices (K, A) and (K, B), respectively. Operator $\mathcal{R}$ denotes a clockwise rotation by $\pi/2$. The idea of this construction approach can be summarized as making a geometric vector decomposition on $\boldsymbol{x}_{K,B} - \boldsymbol{x}_K$ and $\boldsymbol{x}_{K,A} - \boldsymbol{x}_K$ in Λ_K space (Lipnikov, Shashkov, Svyatskiy, & Vassilevski, 2007; Wu & Gao, 2014). The sum of $\alpha_{K,\sigma,A}$ and $\alpha_{K,\sigma,B}$ can be shown to be (Gao & Wu, 2015)

$$\alpha_{K,\sigma,A} + \alpha_{K,\sigma,B} = |\sigma| \frac{\boldsymbol{n}_{K,\sigma}^T \Lambda_K \mathcal{R}(\boldsymbol{x}_{K,B} - \boldsymbol{x}_{K,A})}{(\boldsymbol{x}_{K,A} - \boldsymbol{x}_K)^T \mathcal{R}(\boldsymbol{x}_{K,B} - \boldsymbol{x}_K)} = \frac{|\sigma|}{d_{K,\sigma}} \boldsymbol{n}_{K,\sigma}^T \Lambda_K \boldsymbol{n}_{K,\sigma}, \tag{15}$$

where $d_{K,\sigma}$ represents the distance from point K to edge σ. Since the diffusion tensor Λ_K is required to be positive definite, the condition (12) on coefficients $\alpha_{K,\sigma,A}$ and $\alpha_{K,\sigma,B}$ is satisfied. Similarly, the one-sided flux, $J_{L,\sigma}$, is constructed as

$$J_{L,\sigma} = \alpha_{L,\sigma,A}(f_L - f_{L,A}) + \alpha_{L,\sigma,B}(f_L - f_{L,B}). \tag{16}$$

The coefficients $\alpha_{L,\sigma,A}$ and $\alpha_{L,\sigma,B}$ are calculated in the same way as Equations (13) and (14).

The total flux from cell K to cell L through the edge σ, $\tilde{J}_{K,\sigma}$, is then expressed as a linear combination of $J_{K,\sigma}$ and $J_{L,\sigma}$; i.e.,

$$\tilde{J}_{K,\sigma} = \mu_{K,\sigma} J_{K,\sigma} - \mu_{L,\sigma} J_{L,\sigma}, \tag{17}$$

$$\mu_{K,\sigma} + \mu_{L,\sigma} = 1, \tag{18}$$

where $\mu_{K,\sigma}$ and $\mu_{L,\sigma}$ are two coefficients to be provided below. As discussed above, $\tilde{J}_{K,\sigma}$ is the flux used to update f_K or f_L in the finite volume method. To satisfy the local conservation property of finite volume methods, the flux from cell L to cell K through σ, $\tilde{J}_{L,\sigma}$, is then

$$\tilde{J}_{L,\sigma} = \mu_{L,\sigma} J_{L,\sigma} - \mu_{K,\sigma} J_{K,\sigma}, \tag{19}$$

so that $\tilde{J}_{K,\sigma} + \tilde{J}_{L,\sigma} = 0$.

Based on the NTPFA idea and equations for the one-sided fluxes (9) and (16), we define two new coefficients

$$a_{K,\sigma} = \alpha_{K,\sigma,A} f_{K,A}^t + \alpha_{K,\sigma,B} f_{K,B}^t, \tag{20}$$

$$a_{L,\sigma} = \alpha_{L,\sigma,A} f_{L,A}^t + \alpha_{L,\sigma,B} f_{L,B}^t. \tag{21}$$

Here $f_{K,A}^t$ and $f_{K,B}^t$ are obtained by bilinear interpolation from the four surrounding cell-centered values at time t, since we use orthogonal rectangular meshes. Then we choose $\mu_{K,\sigma}$ and $\mu_{L,\sigma}$ to be

$$\mu_{K,\sigma} = \begin{cases} 0.5 & a_{K,\sigma} = a_{L,\sigma} = 0, \\ \dfrac{|a_{L,\sigma}|}{|a_{K,\sigma}| + |a_{L,\sigma}|} & \text{otherwise}, \end{cases} \tag{22}$$

and $\mu_{L,\sigma} = 1 - \mu_{K,\sigma}$. Set

$$B_\sigma = \begin{cases} \mu_{L,\sigma} a_{L,\sigma} - \mu_{K,\sigma} a_{K,\sigma} & \sigma \in \text{inner cell edges}; \quad \sigma \in C_K \cap C_L \\ -a_{K,\sigma} & \sigma \in \text{Dirichlet boundaries}; \quad \sigma \in C_K \cap \Gamma_D. \end{cases} \tag{23}$$

and

$$B_\sigma^+ = \frac{|B_\sigma| + B_\sigma}{2}, \tag{24}$$

$$B_\sigma^- = \frac{|B_\sigma| - B_\sigma}{2}, \tag{25}$$

then, using coefficients evaluated at time t, we approximate the flux in Equation (17) at time $t + \Delta t$ as

$$\tilde{J}_{K,\sigma} = A_{K,\sigma} f_K^{t+\Delta t} - A_{L,\sigma} f_L^{t+\Delta t} + B_\sigma^\epsilon, \tag{26}$$

where

$$A_{K,\sigma}(f) = \mu_{K,\sigma}(\alpha_{K,\sigma,A} + \alpha_{K,\sigma,B}) + \frac{B_\sigma^+}{f_K^t + \epsilon}, \tag{27}$$

$$A_{L,\sigma}(f) = \mu_{L,\sigma}(\alpha_{L,\sigma,A} + \alpha_{L,\sigma,B}) + \frac{B_\sigma^-}{f_L^t + \epsilon}, \tag{28}$$

$$B_\sigma^\epsilon = \frac{B_\sigma^+ \epsilon}{f_K^t + \epsilon} - \frac{B_\sigma^- \epsilon}{f_L^t + \epsilon}. \tag{29}$$

Here ϵ denotes a small positive number, typically near machine precision. The important thing to note here is that $A_{K,\sigma}$ and $A_{L,\sigma}$ are positive by construction. This property will be used below to prove the positivity preservation of the new method. The term B_σ^ϵ is neglected following Gao and Wu (2015); it vanishes when $a_{K,\sigma} a_{L,\sigma} \geq 0$, but otherwise its neglect may affect accuracy. Thus, we obtain

$$\tilde{J}_{K,\sigma} = A_{K,\sigma} f_K^{t+\Delta t} - A_{L,\sigma} f_L^{t+\Delta t}, \tag{30}$$

and $\tilde{J}_{L,\sigma} = -\tilde{J}_{K,\sigma}$.

When dealing with an edge subject to the Dirichlet boundary condition, we directly use the one-sided flux as $\tilde{J}_{K,\sigma}$; i.e.,

$$\tilde{J}_{K,\sigma_D} = J_{K,\sigma_D} = (\alpha_{K,\sigma,A} + \alpha_{K,\sigma,B}) f_K^{t+\Delta t} + B_\sigma^+ - B_\sigma^-. \tag{31}$$

This can be written as

$$\tilde{J}_{K,\sigma_D} \approx \left(\alpha_{K,\sigma_D,A} + \alpha_{K,\sigma_D,B} + \frac{B_\sigma^+}{f_K^t + \epsilon} \right) f_K^{t+\Delta t} - B_\sigma^-. \tag{32}$$

Clearly, the coefficient is just A_{K,σ_D} with $\mu_{K,\sigma} = 1$.

When dealing with Neumann boundary conditions, σ_N, we directly use the flux:

$$\tilde{J}_{K,\sigma_N} \approx |\sigma_N| q_N. \tag{33}$$

Next, we focus on the K_{th} cell of the simulation domain and apply the above flux estimate to Equation (7).

$$G_K \frac{\Delta C_K}{\Delta t} (f_K^{t+\Delta t} - f_K^t) = -\sum_\sigma \tilde{J}_{K,\sigma} - \sum_{\sigma_D} \tilde{J}_{K,\sigma_D} - \sum_{\sigma_N} \tilde{J}_{K,\sigma_N}. \tag{34}$$

The summation is over all edges of the cell. Substituting the above forms for the flux, we have

$$G_K \frac{\Delta C_K}{\Delta t} (f_K^{t+\Delta t} - f_K^t) = -\sum_\sigma \left(A_{K,\sigma} f_K^{t+\Delta t} - A_{L,\sigma} f_{L,\sigma}^{t+\Delta t} \right) \tag{35}$$

$$- \left(\sum_{\sigma_D} A_{K,\sigma_D} \right) f_K^{t+\Delta t} + \sum_{\sigma_D} B_{\sigma_D}^- - \sum_{\sigma_N} \left[|\sigma_N| q_N \right]. \tag{36}$$

Note that $q_N \leq 0$.

Denoting $U_{KK} = G_K \Delta C_K / \Delta t$, the above equation can be further reorganized as

$$\left(U_{KK} + \sum_\sigma A_{K,\sigma} + \sum_{\sigma_D} A_{K,\sigma_D} \right) f_K^{t+\Delta t} - \sum_\sigma A_{L,\sigma} f_{L,\sigma}^{t+\Delta t} \tag{37}$$

$$= U_{KK} f_K^t + \sum_{\sigma_D} B_{\sigma_D}^- - \sum_{\sigma_N} \left[|\sigma_N| q_N \right]. \tag{38}$$

Rewriting the above equation as

$$\mathsf{M}\boldsymbol{f}^{t+\Delta t} = \boldsymbol{R}. \tag{39}$$

Then

$$R_K = U_{KK} f_K^t + \sum_{\sigma_D} B^-_{\sigma_D} - \sum_{\sigma_N} \left[|\sigma_N| q_N \right]. \tag{40}$$

and

$$M_{KK} = U_{KK} + \sum_{\sigma} A_{K,\sigma} + \sum_{\sigma_D} A_{K,\sigma_D}, \tag{41}$$

$$M_{KL} = -A_{L,\sigma}. \tag{42}$$

Here L is the neighboring cell sharing the edge σ with K.

By construction, $R_K \geq 0$, and M has the following properties $M_{KK} > 0$, $M_{KL} \leq 0$, and the summation along the K_{th} column is positive; i.e.,

$$\sum_L M_{LK} > 0. \tag{43}$$

To see this, note that the off-diagonal contributions cancel the interior-edge terms; therefore,

$$\sum_L M_{LK} = U_{KK} + \sum_{\sigma_D} A_{K,\sigma_D} > 0. \tag{44}$$

These properties make M a nonsingular M-matrix; i.e., M^{-1} is non-negative. Correspondingly,

$$\boldsymbol{f}^{t+\Delta t} = \mathsf{M}^{-1} \cdot \boldsymbol{R} \geq 0. \tag{45}$$

Therefore, the constructed finite volume method is positivity-preserving.

The term $-f/\tau$ can represent either a loss ($\tau_K > 0$) or a source ($\tau_K < 0$). To preserve positivity, we treat losses implicitly and sources explicitly (thanks to Bernard Hass at GFZ for pointing out that the original operator splitting approach may become inaccurate if Δt is large). For $\tau_K > 0$, the loss contribution is

$$-G_K \Delta C_K \frac{f_K^{t+\Delta t}}{\tau_K}, \tag{46}$$

so the diagonal entry becomes

$$M_{KK} = U_{KK} + \sum_{\sigma} A_{K,\sigma} + \sum_{\sigma_D} A_{K,\sigma_D} + G_K \frac{\Delta C_K}{\tau_K} \tag{47}$$

$$= U_{KK} \left(1 + \frac{\Delta t}{\tau_K} \right) + \sum_{\sigma} A_{K,\sigma} + \sum_{\sigma_D} A_{K,\sigma_D}. \tag{48}$$

For $\tau_K < 0$, the source contribution is instead added to the right-hand side:

$$R_K \longrightarrow R_K + G_K \Delta C_K \frac{f_K^t}{|\tau_K|}. \tag{49}$$

Both treatments preserve the nonnegative right-hand side and the M-matrix property, so the method remains positivity-preserving.

References

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